WorldConsistMem: Cross-Query Consistency Evaluation for Long-Term Memory in Evolving Worlds

Anonymous authors
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Abstract

Standard long-term memory evaluations score answers one query at a time. This misses a concrete failure mode: a system can answer individual questions correctly while violating constraints that link those answers into one coherent world history.

WorldConsistMem is a controlled synthetic diagnostic for that failure mode. Dependent query bundles are grounded in one deterministic gold history; machine-checkable constraints $\mathcal{C}$ are derived from that history; a reader conditioned on the memory state is scored by a joint predicate Φ independent of memory internals. Per-query Acc measures exact match to gold; Bundle Consistency Rate (BCR) measures the fraction of bundles with $\Phi = 1$. The gap $\text{Gap}_{\text{query}} = \text{Acc} - \text{BCR}$ is therefore a different evaluation object from Acc, not a monotone transform of it. Constraint Satisfaction Rate (CSR) provides partial diagnostics when BCR floors.

Empirically (frozen Option A: 102 worlds, 1,088 bundles, 7,140 queries), Acc and BCR diverge under corruption and at symbolic scale (e.g., BM25 Acc = 0.929 with BCR = 0.591). Under a local 3B Qwen reader, BCR = 0 for all systems while CSR remains weakly discriminative (0.258–0.329); gold answers on the same subset achieve BCR = 1. Labelled robustness checks (seeds, relation density, distractors, shortcut baselines) preserve large Acc–BCR gaps and are reported separately from the freeze. Architecture superiority is not supported. The benchmark is a diagnostic instrument for synthetic evolving worlds, not a real-world memory proxy.

1 Introduction

Long-term memory systems answer questions about evolving worlds: what is true now, what was true earlier, how transitions occurred, and which evidence supports those claims. Most evaluations score these questions one at a time via answer accuracy, retrieval fitness, conflict handling, or lifecycle metrics (Wu et al., 2025; 2026; Hu et al., 2025; Tao et al., 2026; Xie et al., 2026; Wang et al., 2026; Hu et al., 2026; Liu et al., 2026). Per-query Acc can look strong while the answer set still fails as one coherent world history.

Central failure mode. A system may correctly name the current CEO and the previous CEO, yet assert an impossible transition between them (Figure 2). Each answer can match gold under exact match; the bundle is nevertheless inconsistent. Ordinary Acc does not detect this joint failure.

Research question. Under evolving synthetic worlds with dependent query bundles and machine-checkable cross-query constraints, when do systems that achieve high per-query Acc still fail joint consistency, and which partial diagnostics remain informative when strict consistency floors?

We introduce WorldConsistMem as a controlled diagnostic instrument: dependent bundles grounded in one deterministic gold history; constraints $\mathcal{C}$ derived automatically from that history; consistency scored by predicates Φ over the predicted answer set from a reader conditioned on memory state M_T, without access to memory internals. BCR is the fraction

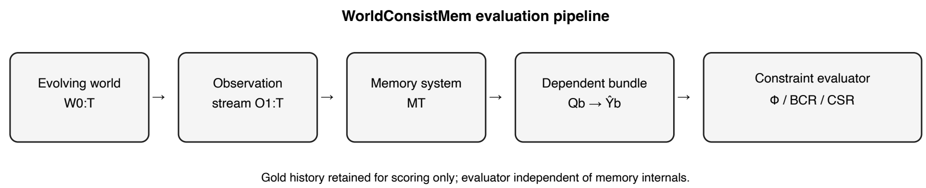

Figure 1: Evaluation object: evolving world → memory → dependent bundle → joint constraint scoring. Takeaway: BCR scores the answer set, not isolated queries.

of bundles with $\Phi = 1$; $\text{Gap}_{\text{query}} = \text{Acc} - \text{BCR}$ measures the Acc–consistency gap; CSR provides partial resolution when BCR floors.

Contributions.

1. Formal evaluation object separating per-query Acc from joint world-history consistency Φ.
2. WorldConsistMem benchmark (102 worlds, 1,088 bundles, 7,140 queries; seed-100 freeze).
3. Metrics BCR, CSR, ConsAcc, and $\text{Gap}_{\text{query}}$, with world-level uncertainty where released.
4. Empirical characterization: corruption and symbolic dissociation; Qwen BCR floor with CSR residual; labelled robustness checks (seeds, density, distractors, shortcut baselines). Architecture superiority is not claimed.

Adjacent logical-consistency work already separates Acc from global coherence on multi-query case files (Salla et al., 2026; Chaudhry, 2026). Existing LTM benchmarks emphasize long-horizon Acc. Our residual is their intersection: persistent memory evaluation over evolving multi-entity worlds with world-history-derived constraints.

2 Related Work

We situate WorldConsistMem against LTM evaluation (Wu et al., 2025; 2026; Hu et al., 2025; Wang et al., 2026; Hu et al., 2026), evolving-knowledge/lifecycle benchmarks (Xie et al., 2026; Liu et al., 2026), conflict/temporal work (Tao et al., 2026), and adjacent cross-query logical consistency (Salla et al., 2026; Chaudhry, 2026). Existing LTM benchmarks primarily emphasize per-query Acc under long histories, updates, conflict, or lifecycle protocols. Adjacent logical-consistency work already studies Acc–global-coherence separation for multi-query reasoning/case files, but not for persistent memory systems over evolving multi-entity gold worlds with world-history-derived constraints.

The residual is not "memory over time" alone, nor "consistency" alone, but their intersection under machine-checkable world-history constraints: dependent bundles, one shared gold history, BCR, and CSR. Detailed comparison appears in Appendix C (Table 6).

3 Cross-Query Consistency in Evolving Worlds

Figure 1 summarizes the evaluation object. Let $W_0, \ldots, W_T$ be structured world states. Events $E_{1:T}$ drive $W_t = \text{Apply}(W_{t-1}, E_t)$. The gold history $(W_{0:T}, E_{1:T})$ is retained for scoring only. The system receives observations $O_{1:T}$ and holds memory state M_T.

A query bundle $Q_b = \{q_1, \ldots, q_{n_b}\}$ asks interdependent current/historical/transition/relational/provenance/conflict questions about one focal lifecycle. Predictions $\hat{Y}_b$ come from a reader on M_T. Constraints $\mathcal{C}_b$ are compiled from the gold history and query slots. Define the joint consistency predicate

$$\Phi(\hat{Y}_b; \mathcal{C}_b) = \mathbf{1}\left[\forall c \in \mathcal{C}_b : \; c(\hat{Y}_b)\right], \tag{1}$$

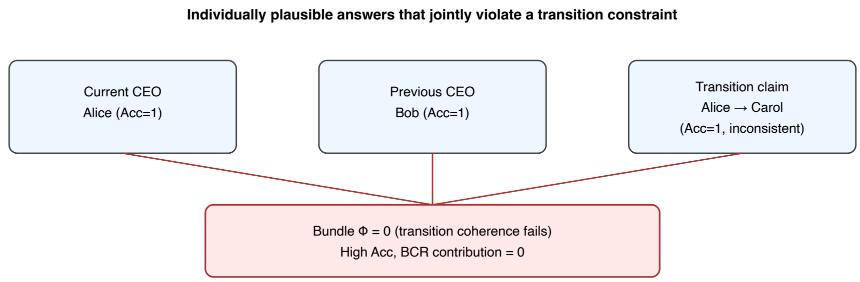

Figure 2: Central failure mode: individually correct answers with an impossible transition ($\Phi = 0$).

Table 1: Dataset statistics (scaled freeze, seed 100). Synthetic by design: controlled gold histories make Φ checkable; ecological validity is not claimed.

Quantity	Value
Worlds (small/medium/large)	102 (34/34/34)
Bundles / queries	1,088 / 7,140
Mean events (S/M/L)	35 / 97 / 215
H4 LLM subset	18 worlds, 192 bundles, 1,260 queries

with $\Phi = 1$ if $\mathcal{C}_b = \emptyset$. Per-query Acc is $\text{Acc}_i = \mathbf{1}[\hat{y}_i \equiv y_i^*]$ under structured equality. Acc aggregates over queries; Φ is a property of the full answer set. Neither is a function of the other: high Acc with $\Phi = 0$ and low Acc with $\Phi = 1$ are both possible (Figure 2; Table 2).

This object is for memory systems over evolving worlds, not SMT satisfiability of logical commitments alone (Salla et al., 2026; Chaudhry, 2026). Any map $O_{1:T} \mapsto M_T$ can be scored with the same $(Q_b, Y_b^*, \mathcal{C}_b)$ protocol.

4 WorldConsistMem Benchmark

WorldConsistMem is a seed-deterministic synthetic benchmark that instantiates Section 3. Worlds contain typed entities and relations; an event grammar emits lifecycle events with validity intervals and provenance IDs. Systems never read gold W_t; they ingest an observation stream only.

For each focal lifecycle the generator emits a dependent bundle spanning current/historical/transition/temporal/provenance/multi-hop/contradiction/counterfactual/conflict queries. Constraints are compiled from gold history and query slots; the evaluator applies $\mathcal{C}_b$ without consulting memory internals. Bundles are not bags of unrelated questions: transition and provenance constraints are meaningful only because answers must agree on one succession.

Table 1 summarizes scale. Tiers differ in event volume (a history-length proxy). Smoke confirms gold Acc=BCR=ConsAcc=1 and Acc–Φ separation under controlled corruptions. The H4 subset (first 6 worlds/tier) is used only for the local Qwen study.

5 Evaluation Metrics

We report metrics jointly. With Φ as in Section 3,

$$\text{Acc} = \frac{1}{N}\sum_i \text{Acc}_i, \qquad \text{BCR} = \frac{1}{|B|}\sum_{b \in B} \Phi(\hat{Y}_b; \mathcal{C}_b), \qquad \text{Gap}_{\text{query}} = \text{Acc} - \text{BCR}, \tag{2}$$

and $\text{CSR}_b = \#\{\text{satisfied } c \in \mathcal{C}_b\}/\#\{\text{applicable } c \in \mathcal{C}_b\}$ (with $\text{CSR}_b = 1$ if $\mathcal{C}_b = \emptyset$). BCR is the fraction of bundles with $\text{CSR}_b = 1$. Acc and BCR have different domains (queries vs.

Table 2: Frozen. Corruption suite: Acc and BCR move independently.

Corruption	Acc	BCR	$\text{Gap}_{\text{query}}$
high_acc_inconsistent	0.849	0.000	0.849
wrong_provenance	0.849	0.000	0.849
consistent_wrong_world	0.394	1.000	−0.606
transition_inconsistency	0.939	0.600	0.339

Table 3: Frozen. Symbolic scaled (102 worlds). Acc/BCR/CSR pooled point estimates (half-up, 3 decimals). Brackets: world-level bootstrap 95% CIs for BCR/CSR.

System	Acc	BCR	$\text{Gap}_{\text{query}}$	mean CSR
B3 recency-k8	0.567	$0.063_{[0.049,0.084]}$	0.504	$0.507_{[0.483,0.547]}$
B4 BM25-k8	0.929	$0.591_{[0.586,0.596]}$	0.338	$0.881_{[0.870,0.890]}$
B1 compact (cap150)	1.000	1.000	0.000	1.000
H0 LTKM (cap150, ret8)	0.976	$0.844_{[0.820,0.898]}$	0.132	$0.973_{[0.970,0.983]}$

bundles) and different predicates (pointwise gold match vs. joint constraint satisfaction); $\text{Gap}_{\text{query}}$ is therefore not a calibrated residual of Acc.

We also report strict ConsAcc, a predeclared PConsAcc grid, family violation rates, abstention/malformed rates for LLM readers, and world-level bootstrap 95% CIs (worlds are the primary units).

6 Evaluated Systems and Setup

Systems are reference implementations for evaluation, not claimed architectural winners. Headline points: capacity-limited compact flat (B1), recency-k8 (B3), BM25-k8 (B4), and a structured lifecycle-aware hierarchical reference H0 (cap150, ret8). Readers: (i) symbolic exact-match over stored/retrieved records (full scaled study); (ii) local Qwen2.5-3B-Instruct-4bit with frozen short-form JSON decoding (temperature 0; max tokens 48) on the H4 subset. Evidence packs are isolated per query under H4 so prompt cross-query leakage is not an uncontrolled confound.

Fairness is imperfect (structure differs; H0 metadata is charged against logical-record budgets; compact padding eviction can retain focal events). Full caveats: Appendix D. Gaps support metric claims; they do not license architectural superiority.

All results below use frozen artefacts unless labelled as a new rescue sanity check. Symbolic and Qwen conditions are never pooled. CSR/PConsAcc are recomputed offline from frozen predictions.

7 Results

Frozen Option A results and labelled NEW robustness checks are never pooled.

7.1 Accuracy and consistency are distinct (frozen)

Corruption (smoke). Gold predictions attain Acc=BCR=1. Controlled corruptions separate the metrics (Table 2).

Symbolic scaled. Retrieval baselines retain large Acc–BCR gaps (Table 3; Figure 3, left). B4 Acc = 0.929 with BCR = 0.591 ($\text{Gap}_{\text{query}}$ = 0.338). Compact flat saturates Acc/BCR at cap150 in this generator regime (fairness caveat: Section 6).

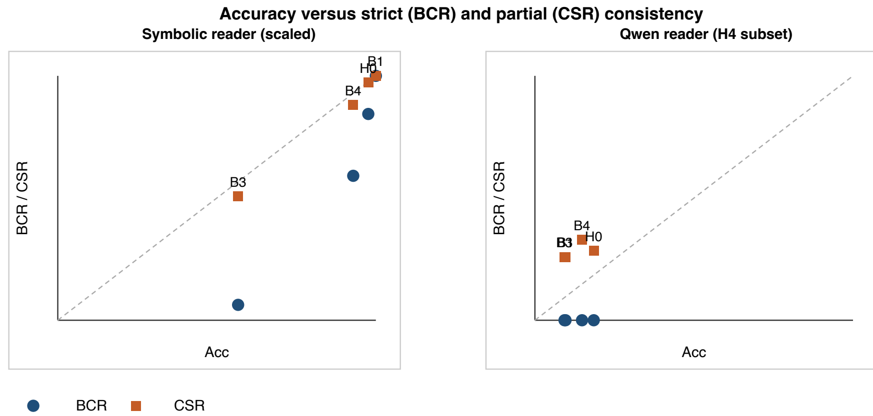

Figure 3: Acc vs BCR/CSR (frozen). Takeaway: Acc can remain high while BCR collapses; when BCR floors, CSR still separates systems.

Table 4: Frozen Option A. Qwen H4 (18 worlds, 192 bundles). All BCR = 0. CSR brackets: world-level bootstrap 95% CIs.

System	Acc	BCR	mean CSR	Abs	Mal
B1 compact	0.096	0.000	$0.258_{[0.239,0.285]}$	0.728	0.007
B3 recency-k8	0.093	0.000	$0.258_{[0.239,0.285]}$	0.728	0.007
B4 BM25-k8	0.148	0.000	$0.329_{[0.314,0.344]}$	0.579	0.001
H0 LTKM	0.185	0.000	$0.284_{[0.275,0.298]}$	0.483	0.029

Component ablations (frozen, symbolic H0). Removing archive or temporal-validity components drops BCR from 0.844 to 0.375 while Acc remains 0.876 ($\text{Gap}_{\text{query}}$ = 0.501); removing graph yields Acc = 0.938, BCR = 0.594 (Table 9). These ablations localize large Acc–BCR gaps to missing lifecycle structure under the symbolic reader; they do not establish architectural superiority under LLM readers.

History-length proxy. Acc–BCR gaps persist across small/medium/large tiers (Figure 5).

7.2 Qwen reader: BCR floors; CSR remains informative (frozen Option A)

On the H4 subset, all four systems have BCR = 0 (Table 4). Acc remains ordered (H0 0.185 highest); CSR separates B4 (0.329) from B1/B3 (0.258) and H0 (0.284). No architecture consistency advantage is supported.

Why BCR = 0 is not a broken metric. Smoke gold and symbolic systems achieve BCR = 1; a labelled H4 gold re-evaluation yields Acc=BCR=1 with zero violations; under Qwen, transition fails on 192/192 bundles with mean 3.33–3.59 violations/bundle (Table 5).

7.3 Robustness and transfer validation (labelled NEW)

The following checks are not Option A and are not merged into frozen tables.

Shortcut / prior baselines. On a downsampled seed-100 protocol (18 worlds), empty answers yield Acc=BCR=0; family-wise gold-marginal majority yields Acc = 0.352 with BCR = 0 ($\text{Gap}_{\text{query}}$ = 0.352); family-shuffle of gold answers yields Acc = 0.280 with

Table 5: Frozen. Qwen violation multiplicity (H4). BCR = 0 reflects multi-constraint failure.

System	Mean viol./bundle	Frac. exactly one	Frac. multiple
B1 / B3	3.59	0.031	0.969
B4 BM25	3.33	0.177	0.823
H0 reference	3.50	0.000	1.000

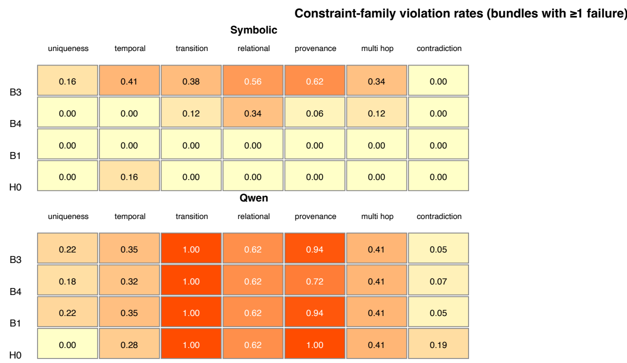

Figure 4: Constraint-family violation rates (frozen). Takeaway: Qwen failures concentrate on transition/provenance.

BCR = 0; gold yields Acc=BCR=1 (Table 10). Non-zero Acc with BCR = 0 under marginal/shuffle controls shows BCR is not a rebranding of Acc.

Seeds. Symbolic B3/B4 at k=8, 6 worlds/tier, seeds 100–105: B3 $\text{Gap}_{\text{query}} = 0.504$ on every seed; B4 $\text{Gap}_{\text{query}} \in [0.335, 0.338]$ (Table 11). Seed diversity is limited under this template-heavy generator (B3 identical; B4 Acc varies only slightly); we report this limitation rather than claim broad seed robustness.

Relation density / distractors. On 8 medium worlds (seed base 200), varying relation density $\{0, 0.2, 0.5, 0.8, 1.0\}$ preserves large B4 gaps ($\text{Gap}_{\text{query}} \in [0.309, 0.317]$) with BCR = 0.600. Toggling distractors on seed-100 with 6 worlds/tier leaves B3/B4 Acc and BCR unchanged under the symbolic reader (null result; Table 12).

Stronger LLM readers. Not completed in this environment (mlx_lm unavailable; limited disk). Option A remains a single 3B Qwen reader.

8 Consistency Failure Analysis

Abstention is high for B1/B3 (0.728), lower for B4 (0.579) and H0 (0.483). Extended tables: Appendix E.

9 Discussion and Limitations

WorldConsistMem isolates a measurable Acc–BCR failure mode for memory systems over evolving synthetic worlds. Completed evidence supports dissociation under corruption, symbolic retrieval, Qwen Option A, and labelled robustness checks.

9.1 Threats to validity

Leakage / shortcuts. LLM evidence packs assert no gold/constraint fields in prompts. Shortcut baselines (Section 7.3) show that answer marginals can yield Acc>0 with BCR = 0;

gold recovers BCR = 1. Symbolic readers use structured match over retrieved observations, not free-form generation.

Generator dependence. Worlds are templated; seed sweeps show limited diversity for B3 under wpt=6. Relation-density overrides preserve gaps but do not strongly change event counts in the tested medium regime. Distractors are a null factor for symbolic B3/B4. Results characterize this generator, not all synthetic worlds.

Model dependence. Option A uses one local 3B Qwen reader with high abstention. Stronger readers were not completed here. Low Acc limits how far Qwen tables speak beyond Acc≠consistency and BCR-floor/CSR questions.

Synthetic scope / sample size. Ecological validity is not claimed. Full symbolic N=102 worlds; Qwen H4 uses 18 worlds; labelled robustness uses smaller protocols and must not be read as Option A.

Fairness. Compact-flat saturation and H0 metadata taxation bound matched-budget interpretation (Appendix D).

9.2 When to use BCR vs CSR

BCR is the strict rate for fully coherent bundles. When BCR floors, report CSR, violation multiplicity, and family hits; do not replace BCR with a favorable PConsAcc cell.

Coverage: Table 13.

10 Conclusion

WorldConsistMem isolates a failure mode in which individual answers can match gold while the answer set violates world-history constraints. Acc and BCR are different evaluation objects; $\mathrm{Gap}_{\mathrm{query}}$ measures their gap. Frozen evidence shows Acc–BCR dissociation under corruption and at symbolic scale; under Qwen Option A, BCR floors while CSR remains weakly discriminative; architecture superiority is unsupported. Labelled robustness checks preserve large gaps and include honest nulls (distractors; limited seed diversity). These conclusions are limited to synthetic worlds, exact-match Acc, the frozen constraint library, and a single 3B-class reader. The contribution is a diagnostic measurement object, not a model ranking or real-world proxy.

AI Use Statement

This statement is required by ICLR 2027 and does not count toward the page limit. We used generative AI coding assistants for manuscript editing, LaTeX packaging, experiment scripting assistance, and repository tooling. We did not use generative AI to invent experimental results, numerical claims, citations, or benchmark labels. All headline numbers were checked against frozen CSVs or labelled re-evaluations. We reviewed AI-assisted text for accuracy and take responsibility for the final content.

Ethics Statement

WorldConsistMem uses fully synthetic worlds, templated observations, and machine-checkable constraints. No human subjects or personal data are involved. Broader societal claims about real-world AI accountability are not made from this synthetic evidence.

Reproducibility Statement

We release seed-deterministic generation (seed 100; 102/1,088/7,140), frozen symbolic and Qwen predictions, metric definitions, decoding freeze, and reference implementations. Partial consistency is recomputed offline from frozen predictions. Labelled robustness outputs (seeds, density, distractors, shortcuts, H4 gold sanity) are stored separately from Option A. Identifying URLs are omitted for double-blind review; anonymized supplementary material accompanies the OpenReview submission. Appendix A details seeds, splits, and artefact checksums.

Acknowledgements

Omitted for double-blind review.

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A Additional Reproducibility Details

Dataset. Scaled WorldConsistMem is generated with seed 100. Smoke validation confirms gold Acc = BCR = ConsAcc = 1 and Acc–Φ separation under controlled corruptions.

Readers. Symbolic exact-match scoring is used for the full scaled study. The LLM condition uses Qwen2.5-3B-Instruct-4bit with frozen short-form JSON decoding (temperature 0; max tokens 48) on the H4 subset (18 worlds / 192 bundles / 1,260 queries).

Systems. Headline reference systems: B1 capacity-limited compact flat; B3 recency retrieval; B4 BM25 retrieval; H0 structured hierarchical reference. Ablations and capacity sweeps are reported in the released artefacts.

Metrics. Primary consistency rate: BCR. Complementary: CSR, ConsAcc / PConsAcc grid, $\text{Gap}_{\text{query}}$, family violation rates. World-level bootstrap 95% CIs accompany BCR/CSR (and world-mean Acc) in frozen CSVs.

Labelled rescue sanity (not Option A). Re-evaluating gold answers on the H4 subset yields Acc=BCR=ConsAcc=1 with zero violations. Additional labelled robustness (seeds 100–105; relation density; distractors; shortcut baselines) is reported in Section 7.3 and Tables 10–12; these artefacts are not merged into Option A tables. Stronger LLM readers were not completed in this environment.

B Artifact Checklist

Released artefacts accompanying this submission include: dataset generators and manifests; symbolic system results and predictions; Qwen H4 system/query-family/constraint-family results; partial-consistency recomputes; corruption-suite metrics; and figure sources corresponding to Figures 1–4. Anonymized supplementary material is provided with the OpenReview submission; identifying repository URLs are omitted from this PDF.

C Extended Related Work

C.1 Long-term memory evaluation

LongMemEval evaluates chat assistants on long-term interactive memory through five ability axes—information extraction, multi-session reasoning, temporal reasoning, knowledge updates, and abstention—with answer accuracy as the primary metric under long-history and long-context protocols (Wu et al., 2025). LongMemEval-V2 extends the agenda toward "experienced colleague" competence, adding dynamic state tracking, workflows, and premise awareness, again with per-question accuracy (and latency-oriented reporting) as the headline outcomes (Wu et al., 2026). MemoryAgentBench evaluates agent memory via incremental multi-turn interactions across competencies such as accurate retrieval, test-time learning, long-range understanding, and selective forgetting / FactConsolidation after edits, scored primarily by task or answer accuracy (Hu et al., 2025). EverMemBench targets long-horizon collaborative, multi-party dialogues and reports fine-grained recall, memory awareness, and profile understanding under QA accuracy protocols (Hu et al., 2026). EvoMemBench organizes agent memory from a self-evolving perspective with a scope$\times$content taxonomy and reports task success or accuracy (and efficiency) across methods compared with long-context baselines (Wang et al., 2026). Taken together, this line provides mature protocols for long-horizon Acc and ability factorization; the measurement gap we target is joint coherence of dependent answers, not another Acc axis.

These benchmarks primarily evaluate individual query or task outcomes. WorldConsistMem does not replace ability Acc suites. It additionally evaluates whether answers within a dependent bundle jointly satisfy constraints derived from one world history. The evaluation unit is the interdependent answer set $\hat{Y}_b$ under $\Phi(\hat{Y}_b; \mathcal{C}_b)$, not a single graded QA item.

Table 6: Closest related benchmarks and adjacent consistency work. Cells use Y / P / N for whether the feature is a primary evaluation object as operationalized in WorldConsistMem.

Work	LTM	Evol.	Bundles	Gold hist.	Trans.	Prov.	CQ constr.	Strict BCR	Partial CSR	Primary metric
LongMemEval	Y	P	N	P	P	N	N	N	N	Ability Acc
LongMemEval-V2	Y	Y	N	P	P	N	N	N	N	Acc (+ latency)
MemoryAgentBench	Y	P	N	P	P	N	N	N	N	Competency Acc
EverMemBench	Y	Y	N	P	P	N	N	N	N	Recall / profile Acc
EvoMemBench	Y	P	N	P	P	N	N	N	N	Task Acc
MemConflict	Y	P	N	P	P	P	P	N	P	Conflict Acc + retrieval
DynamicMem	Y	Y	P	Y	P	N	P	N	P	Checkpoint Acc
WorldMemArena	Y	Y	P	P	P	P	N	N	P	Lifecycle Acc
SetCons metrics	N	N	Y	P	P	N	Y	Y	P	SetCons / satisfiability
LogicVault	N	N	Y	P	P	N	Y	Y	P	Logical / SMT
WorldConsistMem	Y	Y	Y	Y	Y	Y	Y	Y	Y	Acc+BCR+CSR

C.2 Dynamic and evolving knowledge

Evolving worlds are not novel by themselves. DynamicMem constructs long-horizon personal memory in real-world settings and evaluates temporal checkpoint reconstruction and related state-tracking accuracy for evolving user attributes, habits, and preferences (Xie et al., 2026). WorldMemArena evaluates multimodal agent memory through action–world interaction, diagnosing write, maintain, retrieve, and use stages under lifelong evolution and agentic execution protocols (Liu et al., 2026).

It is important to distinguish three notions that are easy to conflate. Tracking changing state asks whether the system recovers the correct value at a time or checkpoint. Profile or lifecycle consistency asks whether personal attributes, stage metrics, or write/use pipelines remain coherent for a user or agent trajectory. Cross-query world-history consistency asks whether interdependent answers about current state, historical state, transitions, relations, and provenance jointly satisfy constraints induced by one shared multi-entity gold history.

C.3 Conflict, temporal validity, and provenance

MemConflict evaluates long-term memory systems under dynamic, static, and conditional memory conflicts, combining black-box answer correctness with white-box retrieval and ranking diagnostics, and showing that answer correctness can diverge from retrieval fitness (Tao et al., 2026). These lines diagnose conflict handling, version validity, and temporal updates at the query, retrieval, or competency level. The WorldConsistMem contrast is evaluation grain and joint structure rather than denial of those pathologies: a conflict- or update-correct single answer can still participate in an inconsistent bundle.

C.4 Cross-query logical consistency

Salla et al. (2026) study case-file logical consistency across interdependent queries and introduce set-level metrics including Case Satisfiability and SetConsRate. Chaudhry (2026) maintain persistent symbolic belief states verified with SMT-style checking and release LogicBench-Cross. Machine-checkable constraint evaluation over multi-query answer sets is therefore not universally new. WorldConsistMem's residual is application domain and constraint origin: persistent memory systems over evolving multi-entity worlds with world-history-derived constraints.

D Fairness Caveats

Comparisons are informative but not perfectly matched:

- Internal structure differs (flat list vs multi-store H0).
- Capacity accounting charges H0 metadata (graph, provenance, indices) against the same logical-record budgets used for flat stores; absolute budgets therefore tax structured metadata.
- Retrieval budgets (k vs store-routed packs) are matched numerically where stated, but lexical BM25 is not identical to H0's routing.

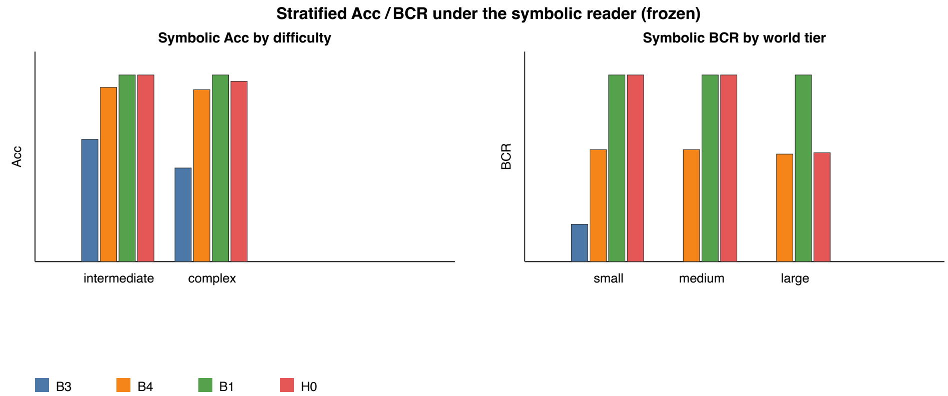

Figure 5: Symbolic Acc by difficulty and BCR by world tier.

Table 7: Constraint-family violation rates (headline systems).

	Symbolic				Qwen		
Family	B3	B4	B1	H0	B1/B3	B4	H0
uniqueness	0.156	0.000	0.000	0.000	0.219	0.177	0.000
temporal	0.406	0.000	0.000	0.156	0.354	0.323	0.281
transition	0.375	0.120	0.000	0.000	1.000	1.000	1.000
relational	0.562	0.343	0.000	0.000	0.625	0.625	0.625
provenance	0.625	0.057	0.000	0.000	0.938	0.724	1.000
multi_hop	0.344	0.124	0.000	0.000	0.406	0.406	0.406
contradiction	0.000	0.000	0.000	0.000	0.052	0.073	0.188

- Deterministic vs LLM readers are separate conditions; symbolic Acc/BCR must not be pooled with Qwen Acc/BCR.
- Compact flat can retain focal lifecycle events by dropping padding, which can yield very high symbolic Acc/BCR at medium budgets; this is a fairness caveat for matched-budget interpretation, not a general ranking of store designs.

E Additional Figures and Tables

Table 8: Query-family Acc under Qwen (H4; frozen; three-decimal display).

Family	B1	B3	B4	H0
current_state	0.010	0.010	0.083	0.542
historical_state	0.162	0.162	0.089	0.547
transition	0.000	0.000	0.000	0.000
multi_hop	0.000	0.000	0.052	0.000
provenance	0.188	0.167	0.417	0.000
temporal_order	0.128	0.128	0.205	0.308
contradiction	0.083	0.083	0.167	0.000
counterfactual	0.000	0.000	0.000	0.000
conflict_validity	0.462	0.462	0.462	0.000

Table 9: Frozen. Symbolic H0 component ablations (reprinted from freeze CSV). Large Acc–BCR gaps under archive/temporal/graph removals.

System	Acc	BCR	$\text{Gap}_{\text{query}}$
H0 unconstrained	1.000	1.000	0.000
H0 cap150 ret8	0.976	0.844	0.132
H–archive	0.876	0.375	0.501
H–temporal-validity	0.876	0.375	0.501
H–graph	0.938	0.594	0.344
H–provenance	1.000	1.000	0.000
H–conflict	0.800	0.812	−0.012

Table 10: Labelled NEW. Shortcut baselines (seed 100; 6 worlds/tier; symbolic scoring). Family-majority uses oracle gold marginals by query family.

Baseline	Acc	BCR	$\text{Gap}_{\text{query}}$
Empty answers	0.000	0.000	0.000
Family-majority (gold marginals)	0.352	0.000	0.352
Family-shuffle of gold answers	0.280	0.000	0.280
Gold oracle	1.000	1.000	0.000

Table 11: Labelled NEW. Seed sweep summary (symbolic B3/B4 k=8; 6 worlds/tier; seeds 100–105). B3 identical across seeds (limited diversity).

System	Acc range	BCR	$\text{Gap}_{\text{query}}$ range
B3 recency-k8	0.567–0.567	0.0625	0.504–0.504
B4 BM25-k8	0.929–0.932	0.5938	0.335–0.338

Table 12: Labelled NEW. Generator controls (symbolic). Density: 8 medium worlds, seed base 200. Distractors: seed 100, wpt=6.

Control	Setting	System	Acc	BCR
Relation density	0.0 / 1.0	B4	0.917 / 0.909	0.600 / 0.600
Relation density	0.0 / 1.0	B3	0.545 / 0.545	0.000 / 0.000
Distractors	on / off	B4	0.932 / 0.932	0.594 / 0.594
Distractors	on / off	B3	0.567 / 0.567	0.062 / 0.062

Table 13: Evaluation coverage and hard limits.

Dimension	Coverage
Paper shape	Benchmark + evaluation methodology
Symbolic reader	Full scaled (102 / 1,088 / 7,140)
LLM reader	Qwen2.5-3B-Instruct-4bit; H4 subset only
Stronger LLM readers	Not completed in this submission environment
Architecture superiority	Unsupported under Qwen (all BCR = 0)
Ecological validity	Not claimed (synthetic worlds)
Primary consistency metric	BCR (strict); CSR complementary
Labelled robustness	Seeds/density/distractors/shortcuts (separate tables)